



Econometrics II

HW 4: Selection Bias - Causality

Due: Friday, Ordibehesht 14, 23:59

Instructions

- Please name your file `lastname.firstname.HW#.zip`.
- Email your work to airanpour22@gmail.com and unesrom@gmail.com before the deadline.
- Submit your work in a single R Markdown file (`.Rmd`), including the HTML file generated by knitting this document.

Part I (Sample Balance) (45 points):

This problem asks you to replicate a part of Krueger's (1999) balance check exercise which you have already seen in the slides in the class. This paper uses the *STAR* dataset which you can find using *AER* package in R (You can simply use this command: `data("STAR", package = "AER")`).

1. Replicate Table 1 of this paper and interpret the results. Are all the samples balanced?
2. Based on the identification strategy utilized in this paper, draw a simple DAG for major channels of effect.

Part II (Average Treatment Effect) (30 points):

In this question, you have the data for a sample of 100 hospitalized patients. There is a surgery intervention, $D = 1$ and a chemotherapy intervention, $D = 0$. Each patient has an observed outcome that is defined as post-treatment life span in years. (See `Surgery.dta`)

1. Write formula for calculating the observed effect of surgery then calculate Observed Effect. Give one reason why this could be misleading in evaluating the surgery intervention.
2. You have the counterfactual outcomes in the dataset. According to this, have specialists in the hospital correctly determined the treatment (surgery or chemo) for all patients?
3. Calculate the Average Treatment Effect (ATE).
4. Calculate average treatment effect on the treated (ATT), as well as average treatment effect on the untreated (ATU). Show that ATE is the weighted average of ATT and ATU using our sample.
5. Now suppose we did not have data on counterfactual outcomes. With what assumptions could we get ATE from the observed effect?

Part III (Variables to Control) (25 points):

The following items describe situations where researchers use the Conditional Independence Assumption (CIA) as their identification assumption. For each scenario, assess their decision on variables to control by attributing each item to one of the 5 cases discussed in class. Include any good controls that help the researcher to capture the causal effect. Also draw the DAG in R.

1. Nutrition Study: A nutritionist wants to determine if consuming a high-protein diet improves cognitive function in adolescents. In her analysis, she controls for the participants' hours of sleep and type of school they attend.
2. Urban Development Study: An urban planner investigates the impact of public parks on neighborhood crime rates. In his model, he includes the average income of the neighborhood and the distance to the nearest police station as control variables.
3. To examine the effect of gender discrimination on women's earnings, a researcher controls for their work experience and IQ in a sample of working women and men.
4. Environmental Study: A researcher explores whether proximity to industrial areas affects respiratory health. In the analysis, she controls for smoking habits.

5. Economic Inequality Study: An economist examines how minimum wage increases affect employment levels. He controls for the overall economic growth rate of the region in his analysis.